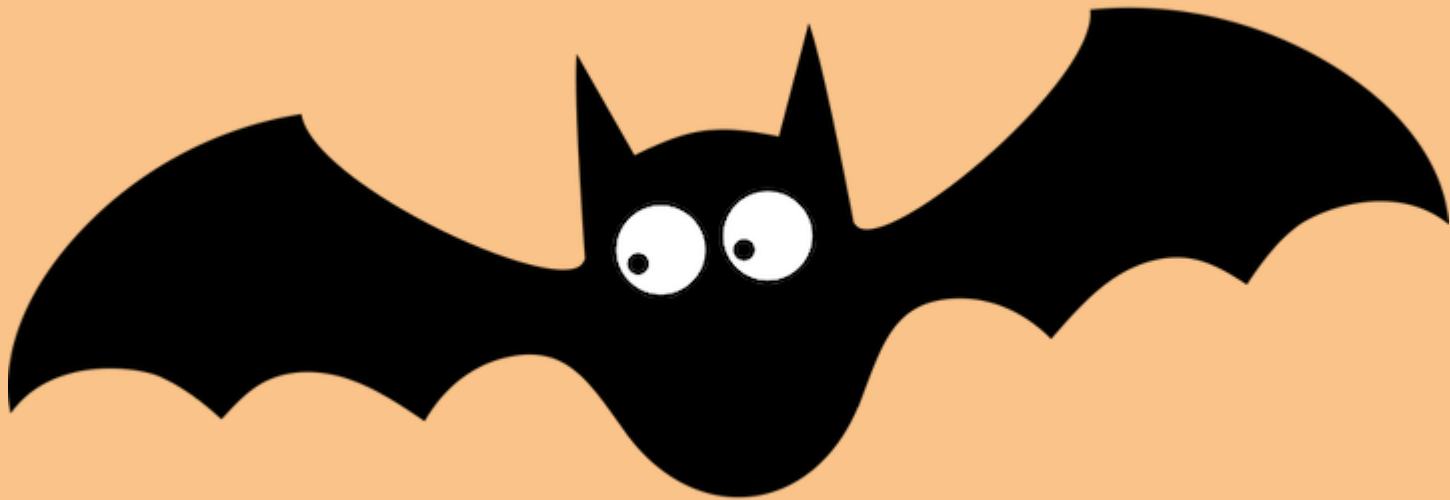




Deep Learning

Son Nguyen

Agenda

- Deep learning revolution
- Python Example
 - Apple diseases Detection: CNN vs. Gradient Boosting
 - Pneumonia X-ray detection
 - GAN
- Define neural networks

Deep Learning

ARTIFICIAL INTELLIGENCE

Any technique that enables computers to mimic human behavior



MACHINE LEARNING

Ability to learn without explicitly being programmed



DEEP LEARNING

Extract patterns from data using neural networks

3 1 3 4 7 2
1 7 4 2 3 5

Example

<https://www.youtube.com/watch?v=l82PxsKHxYc&t=47s>

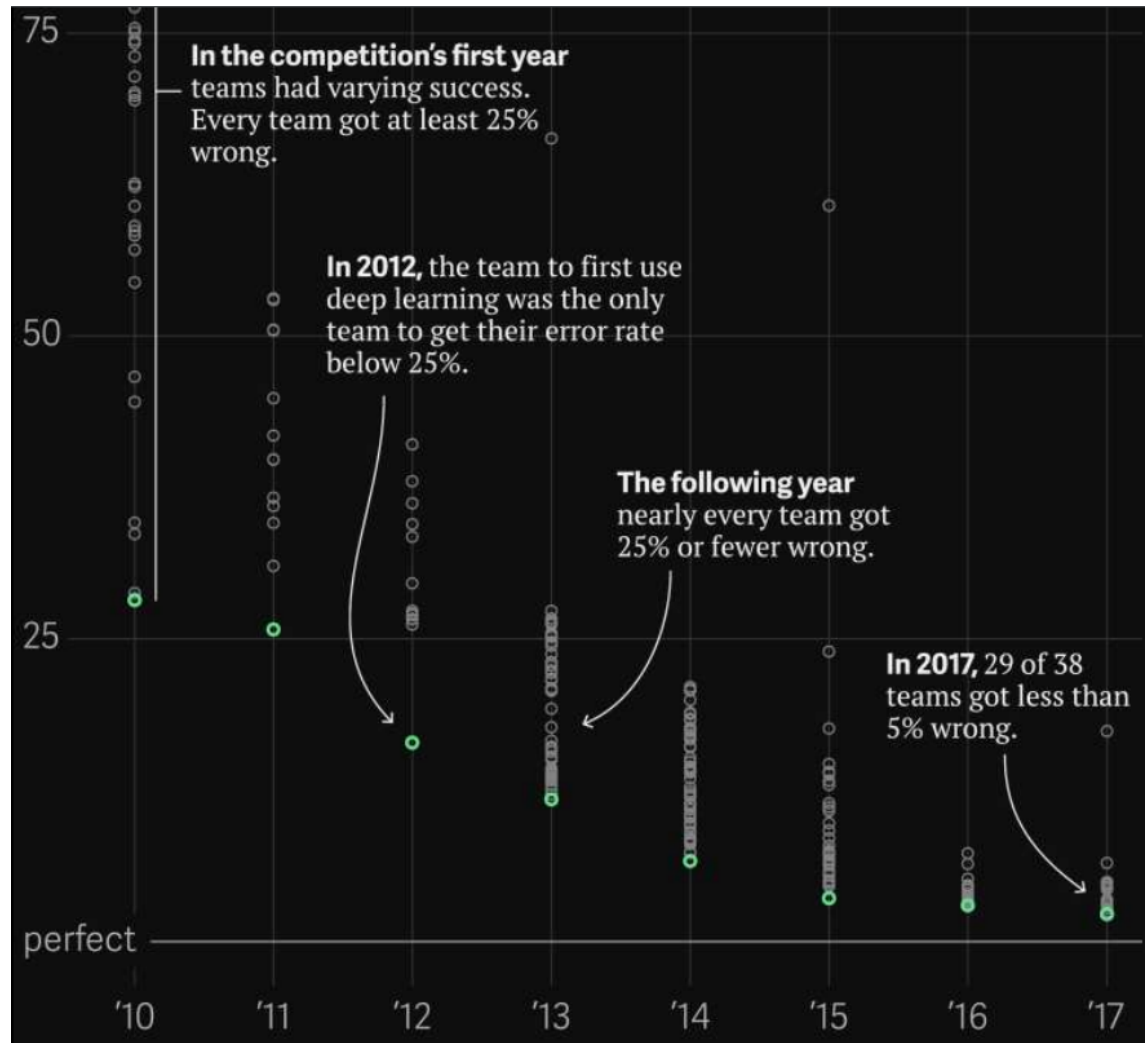
2009: ImageNet Competition

- Launched in 2009
- 15 million high-resolution images
- 22,000 categories



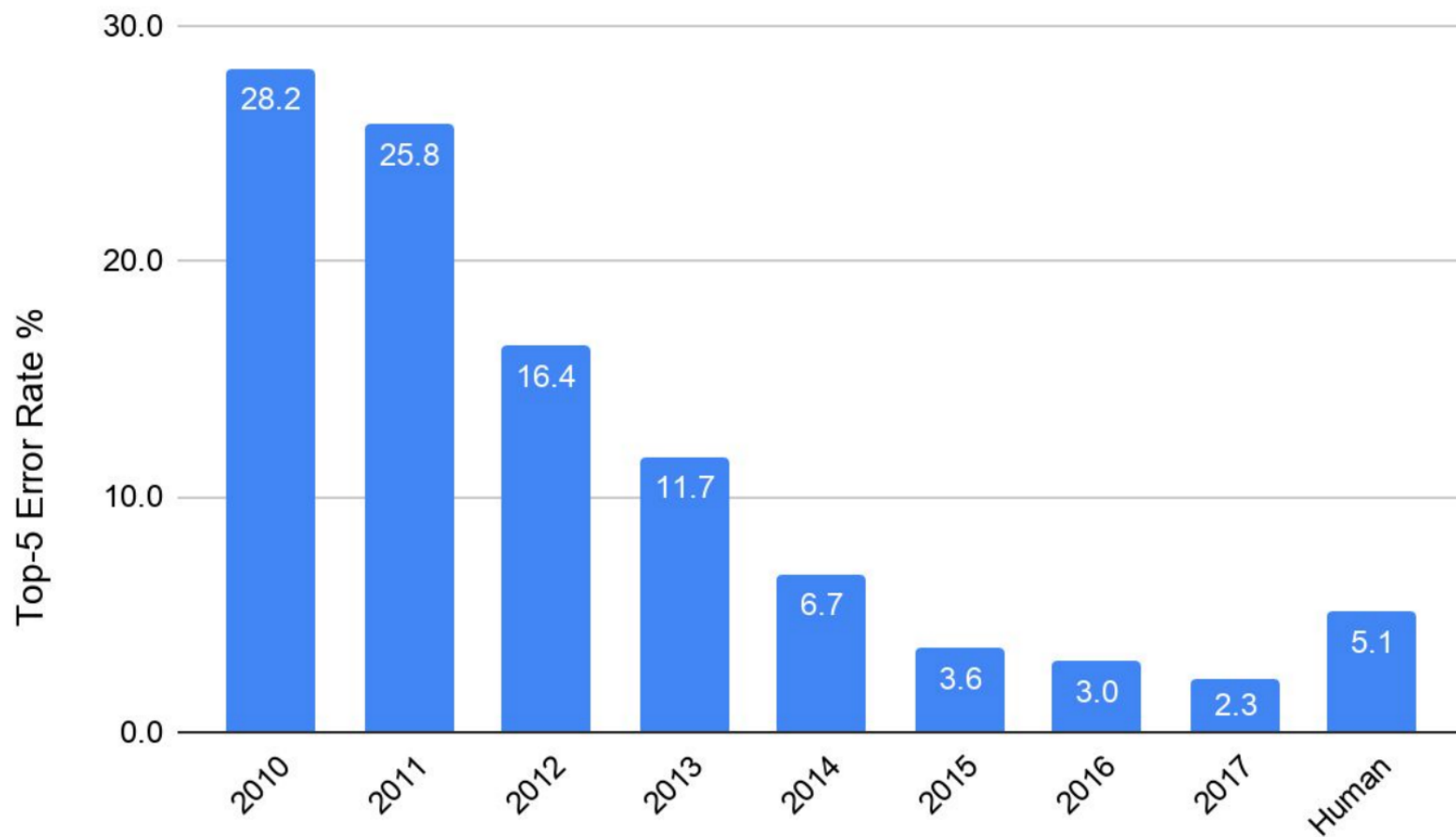
AlexNet

- Dramatic performance gain
- No Teams get better than 25% accuracy by 2012
- AlexNet was introduced in 2012 with 15.3% accuracy
- No team did worse than 25% since

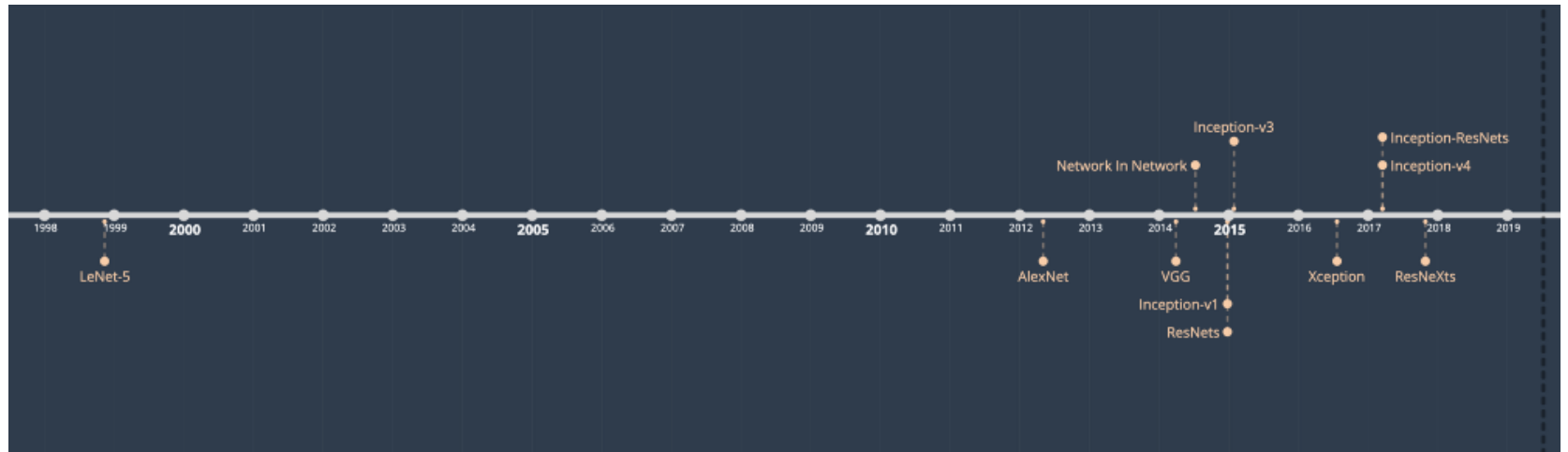


AlexNet

ImageNet Contest Winning Entry Error Rate

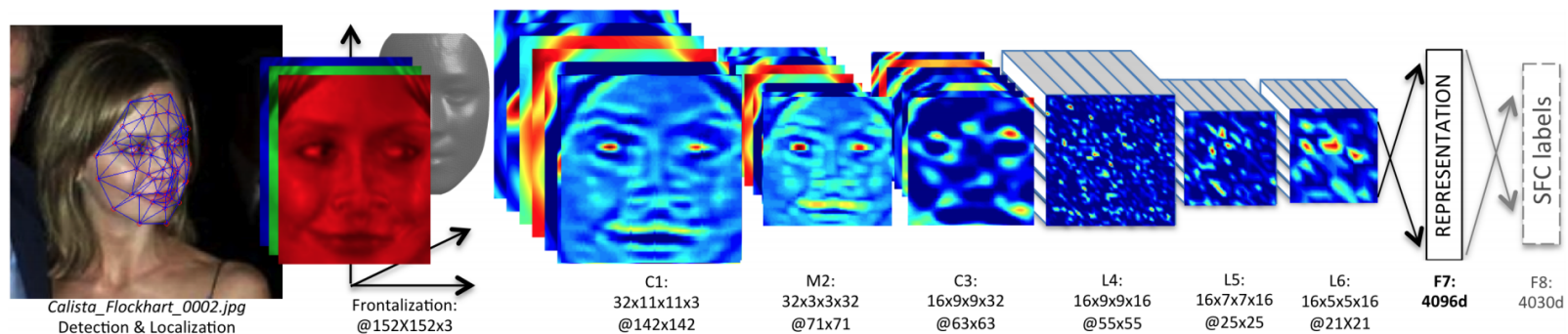


CNN



2014: DeepFace - Facebook

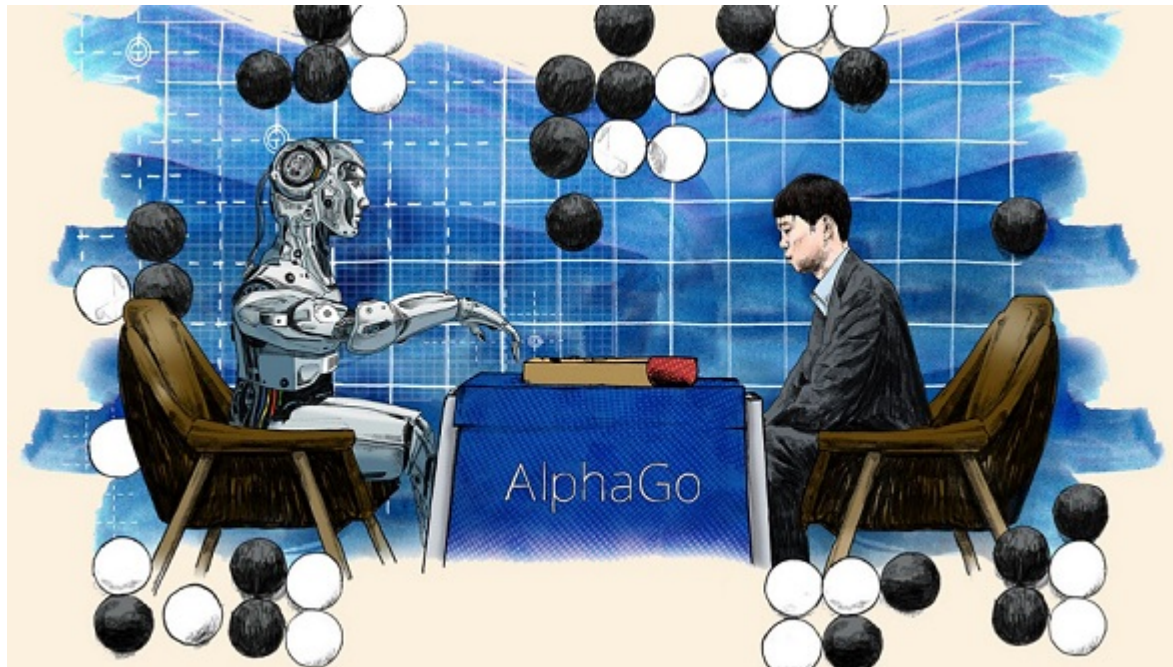
- In 2014, Facebook - DeepFace can identify faces with 97.35% accuracy, beating rivals human performance



Outline of the *DeepFace* architecture.

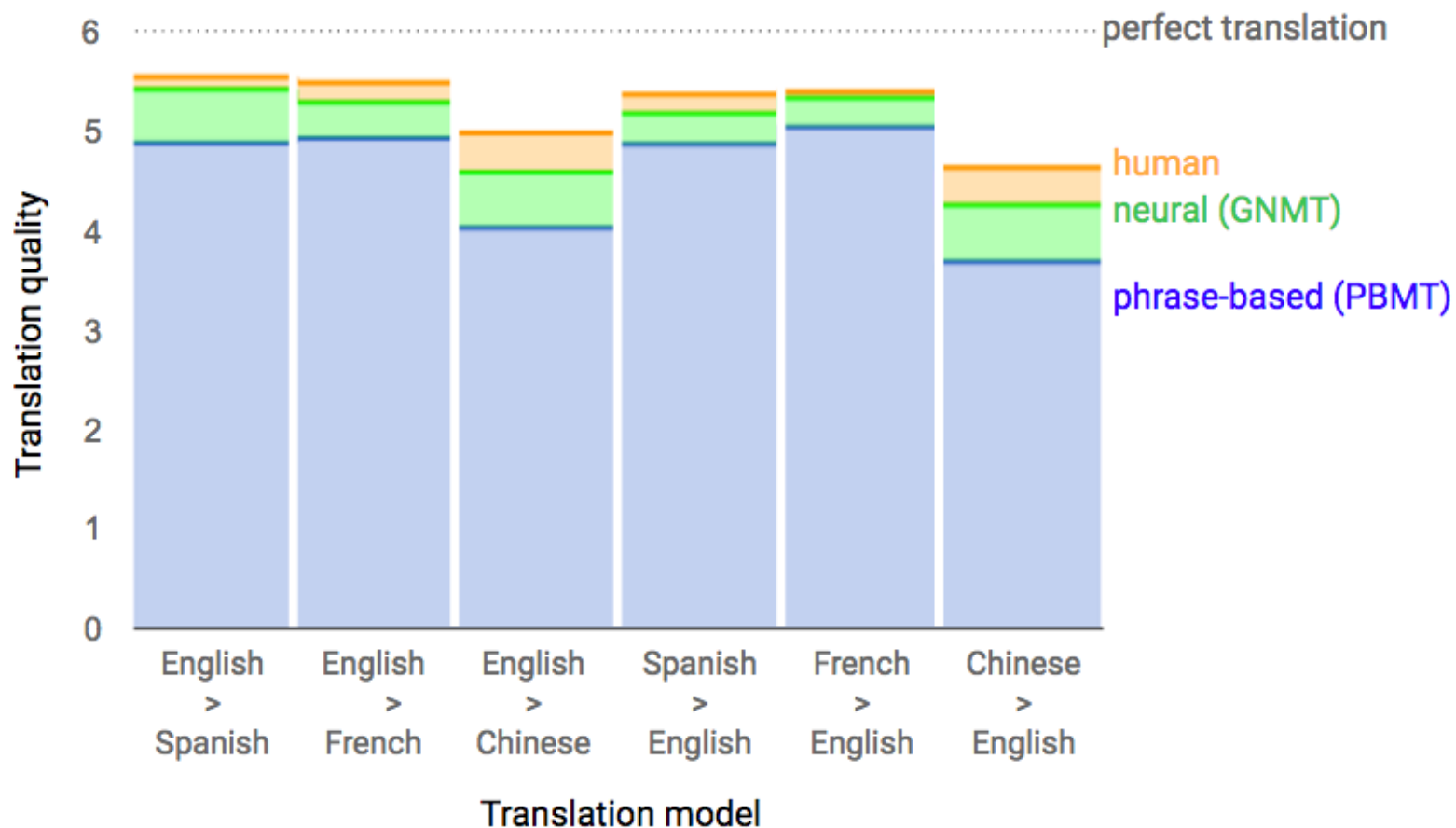
2016: AlphaGo - Google

- In 2016, AlphaGo became the first computer program to defeat a professional human Go player
- the first to defeat a Go world champion,
- arguably the strongest Go player in history.
- AlphaGo was one of the Breakthrough of the Year by Science.



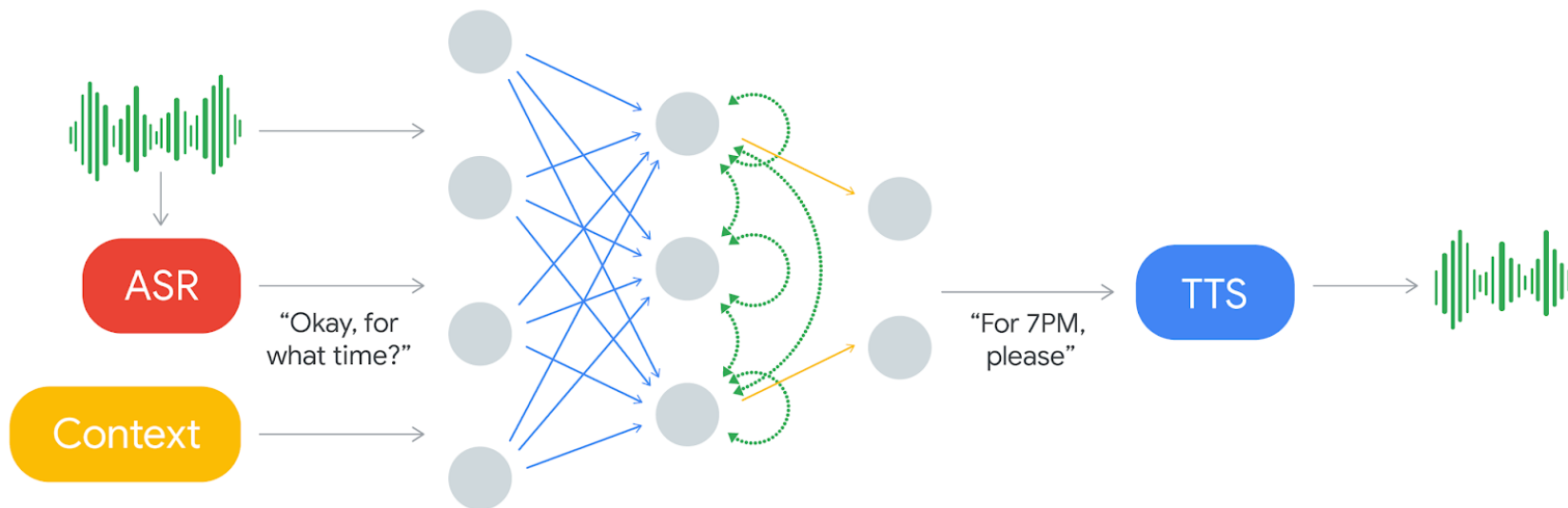
2016: Google Translate

- In 2016, Google Translate Deep Learning blew other language translation techniques out of the water



Google Assistant

<https://ai.googleblog.com/2018/05/duplex-ai-system-for-natural-conversation.html>



2017: Face ID - Apple

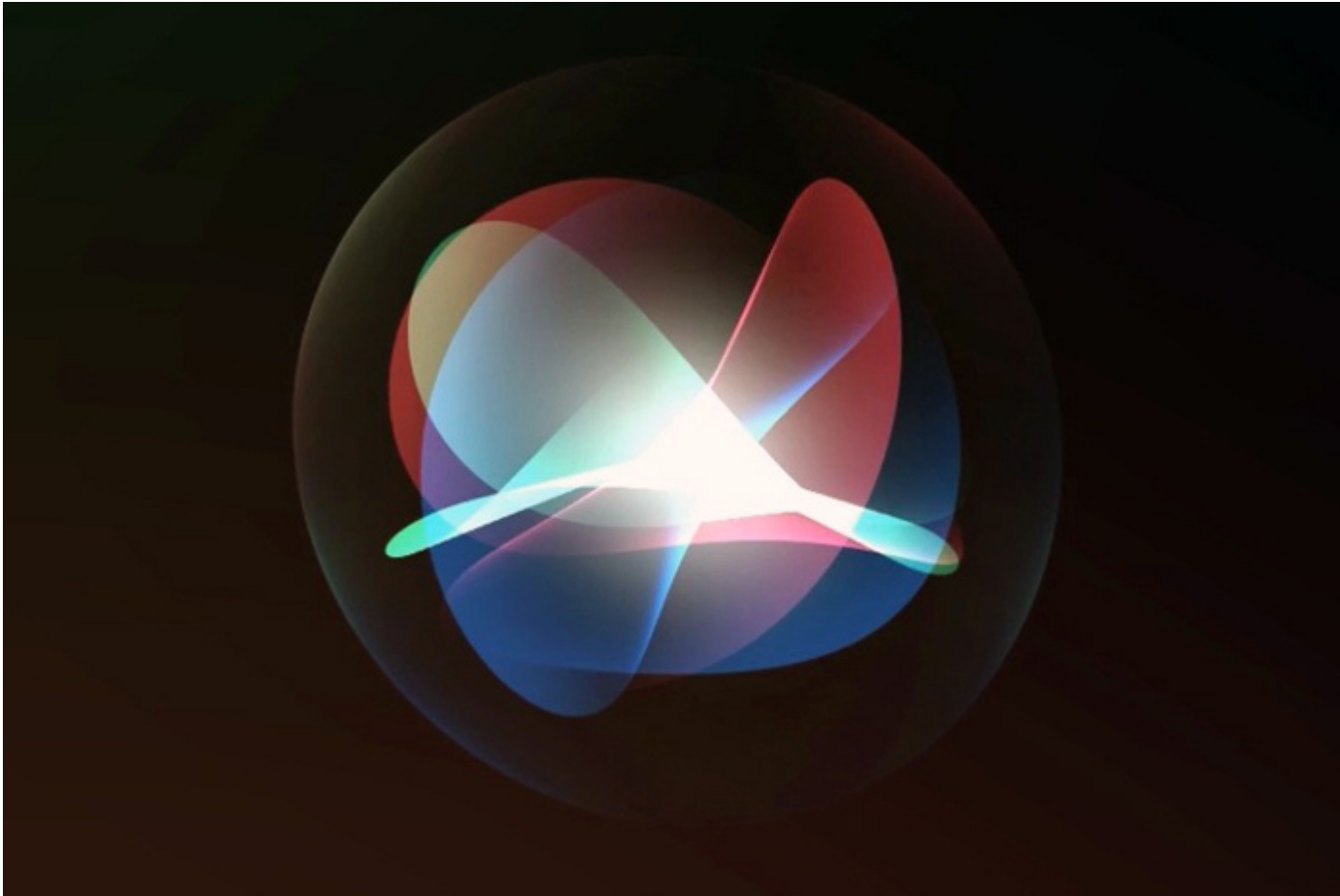
- In 2017, Apple switched to use Deep Learning for face recognition

We had to completely rethink our approach so that we could take advantage of this paradigm shift. <https://machinelearning.apple.com/research/face-detection>

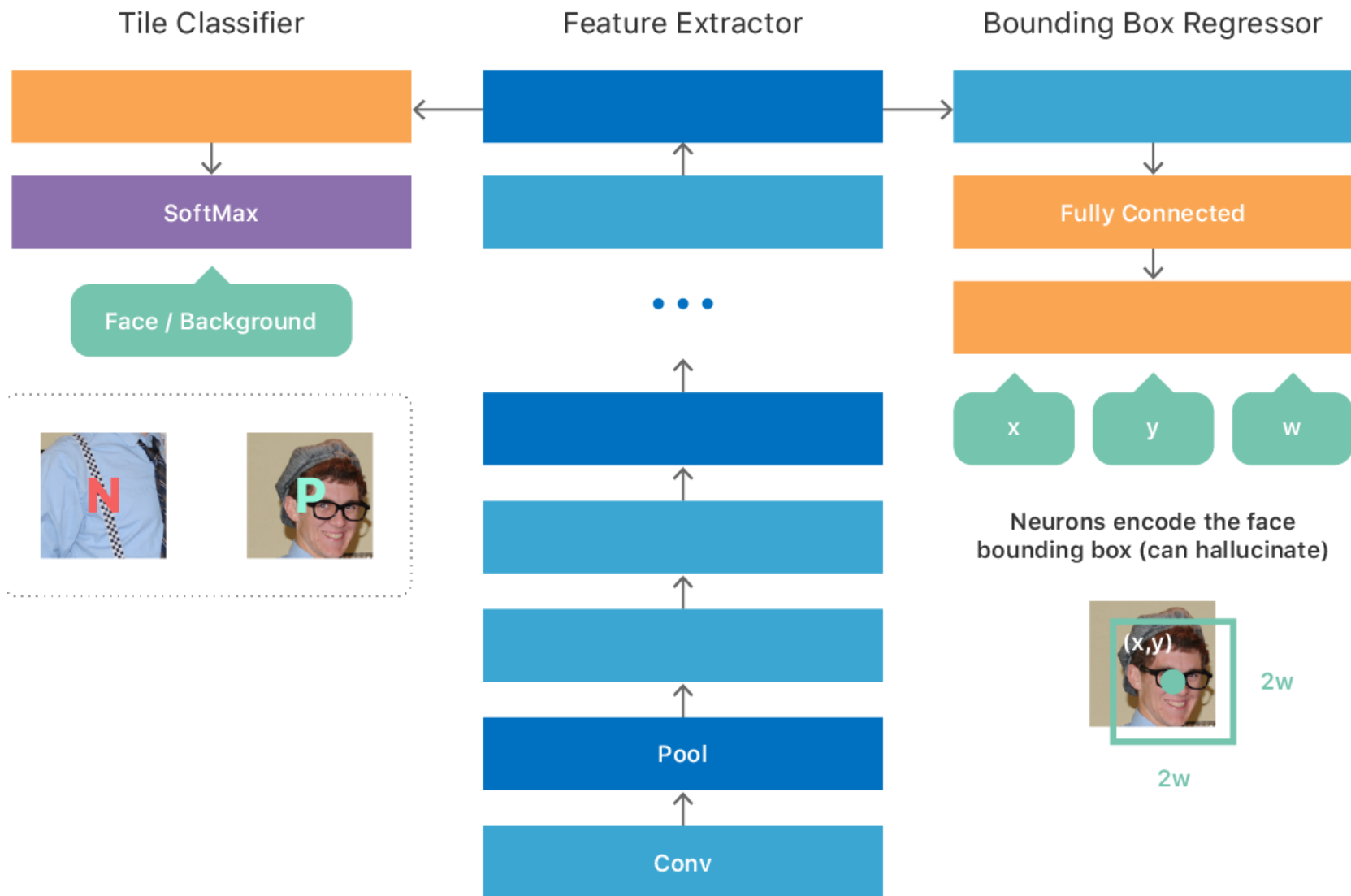


Apple Siri

<https://machinelearning.apple.com/research/siri-voices>

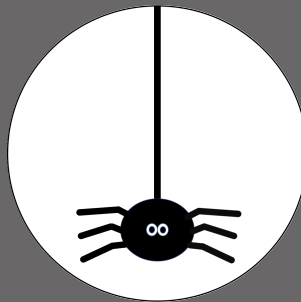


Apple - Deep Learning



Amazon

Academia

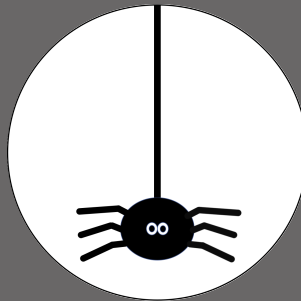


Academia

2018: Alan Turing Award



Big Tech Companies Interested



- In 2014, Google acquired DeepMind.
- <https://deepmind.com/research/case-studies/alphago-the-story-so-far>
- <https://en.wikipedia.org/wiki/DeepMind>
- <https://www.tensorflow.org/about>

Tensorflow

- Google developed TensorFlow in 2015
- particularly focuses on training and inference of deep neural networks.
- Company uses Tensorflow



- <https://www.tensorflow.org/about>

Google Brain

- Google Brain is a deep learning artificial intelligence research team at Google.
- Formed in the early 2010s
- <https://research.google/teams/brain/>

Apple

- Deep Learning Team:

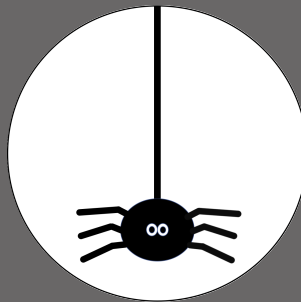
<https://www.apple.com/jobs/us/teams/machine-learning-and-ai.html>

Facebook: Pytorch

- In 2016, Facebook developed PyTorch, an open source deep learning framework

Deep Learning Group: <https://www.microsoft.com/en-us/research/group/deep-learning-group/>

But...

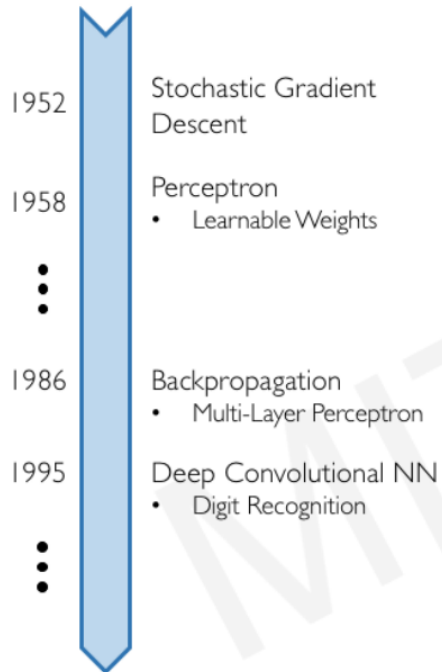


But...

- Neural Network was invented a long time ago
- Why now?

Why Now?

Neural Networks date back decades, so why the resurgence?



1. Big Data

- Larger Datasets
- Easier Collection & Storage

IMAGENET



2. Hardware

- Graphics Processing Units (GPUs)
- Massively Parallelizable

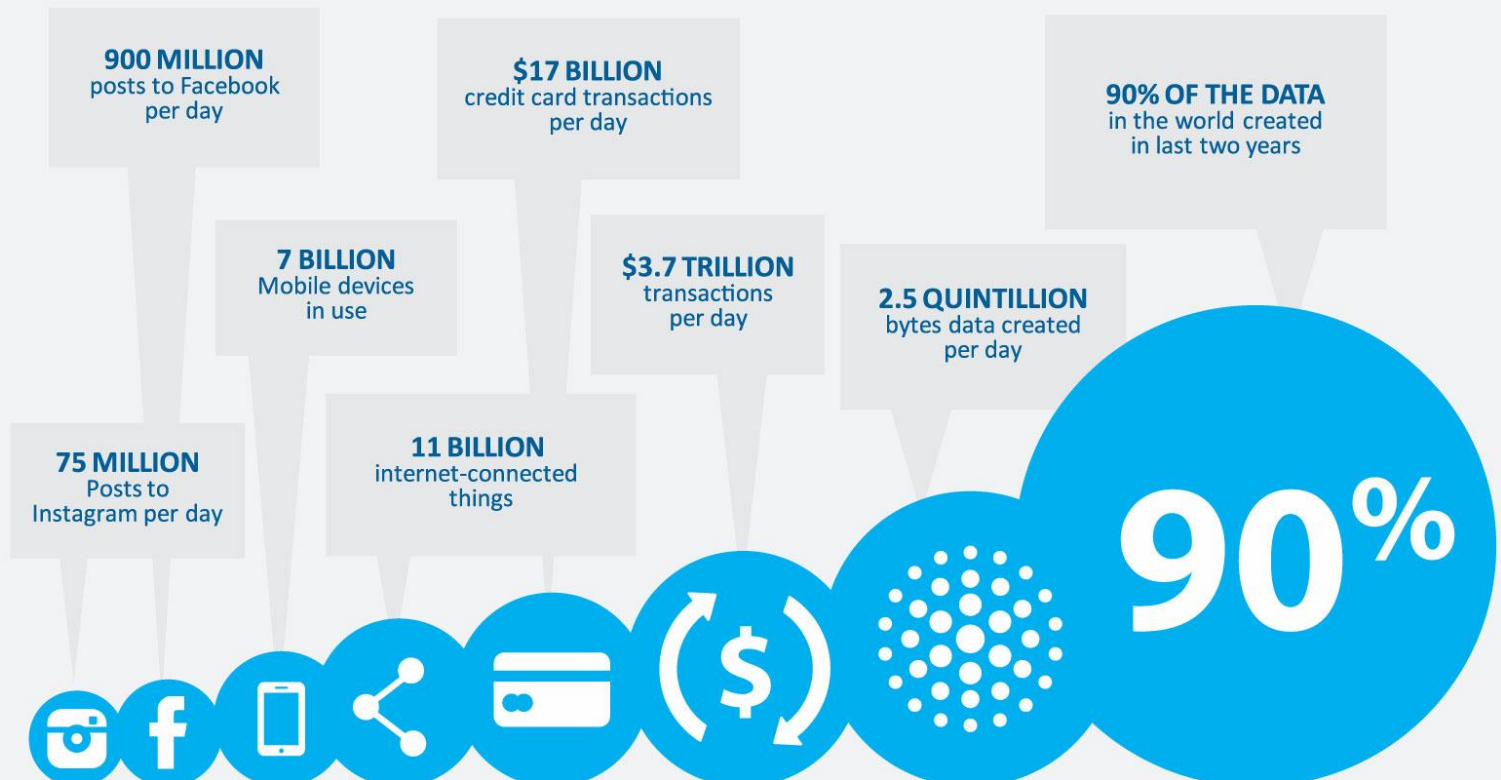


3. Software

- Improved Techniques
- New Models
- Toolboxes

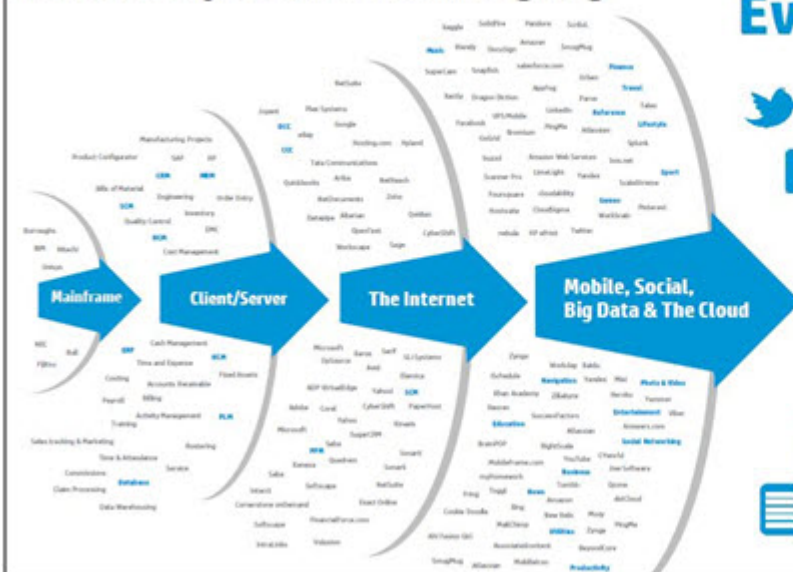


Big Data



Big Data

A new style of IT emerging



Every 60 seconds



98,000+ tweets



695,000 status updates



11million instant messages



698,445 Google searches



168 million+ emails sent

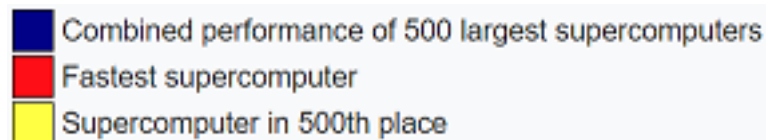
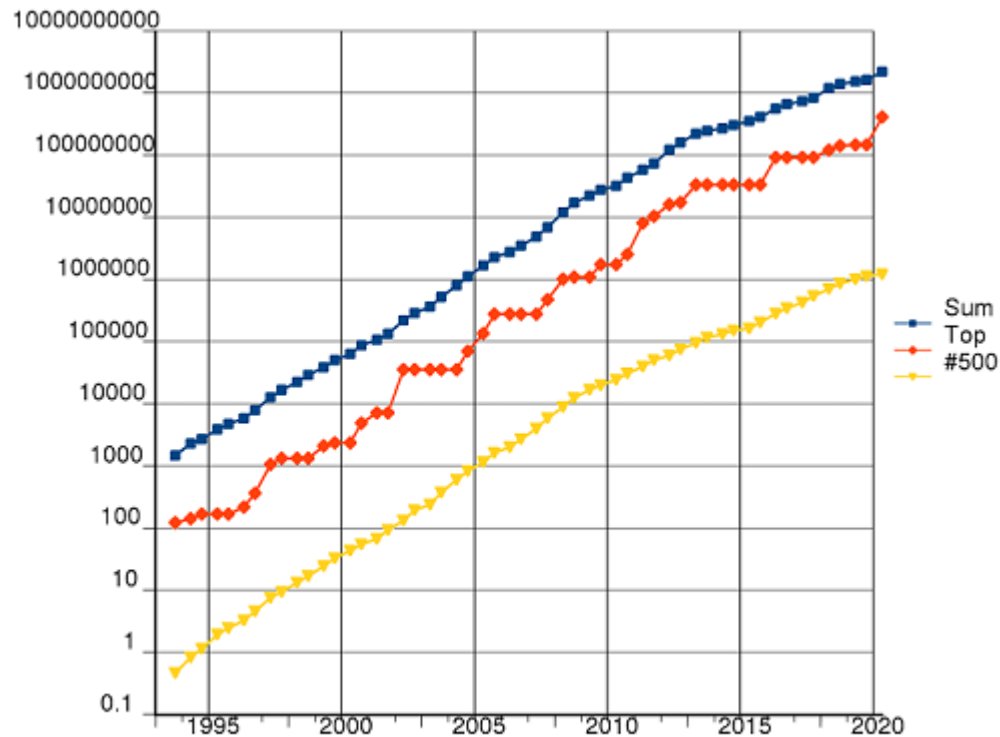


1,820TB of data created



217 new mobile web users

Powerful Computing



- https://en.wikipedia.org/wiki/History_of_supercomputing

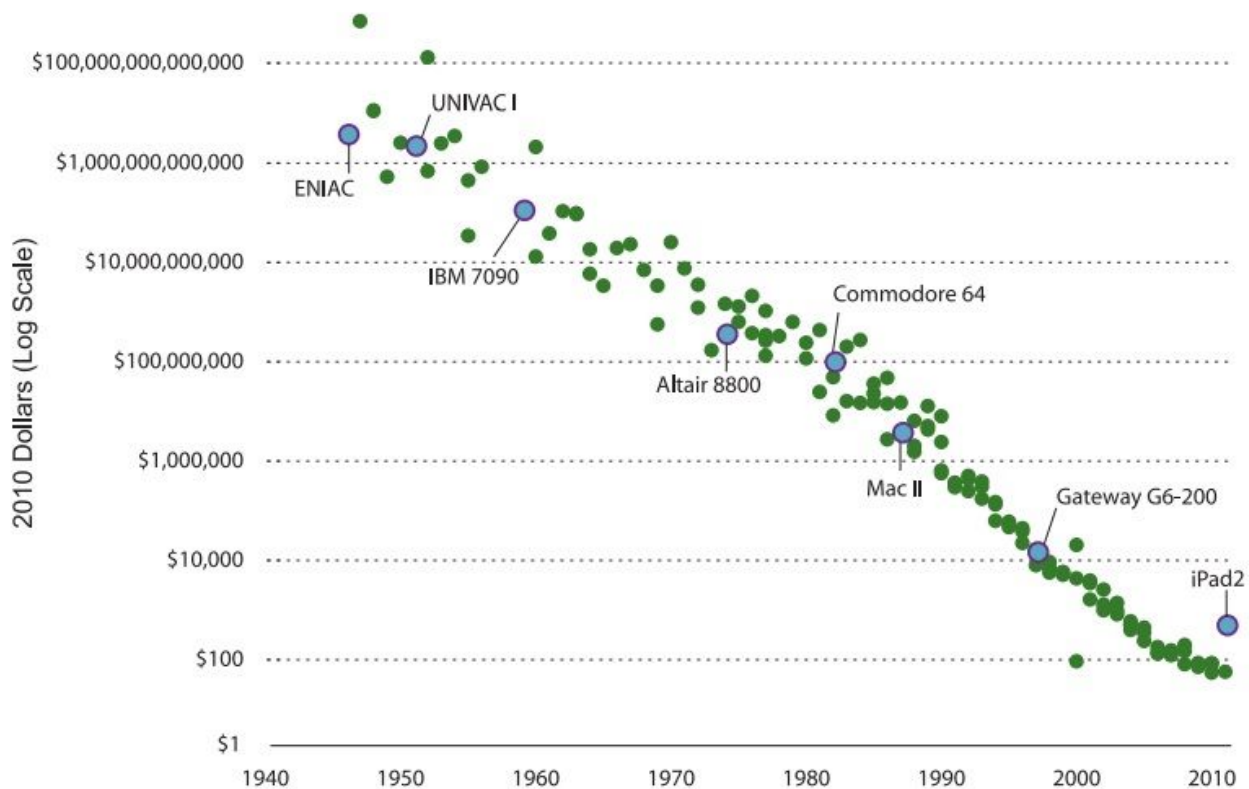
Powerful Computing

Powerful Computing

- If AlphaGoZero was trained in 1 day in 2020, how long would it take to train AlphaGoZero in 2012?
- More than 821 years!

Cheaper Computing

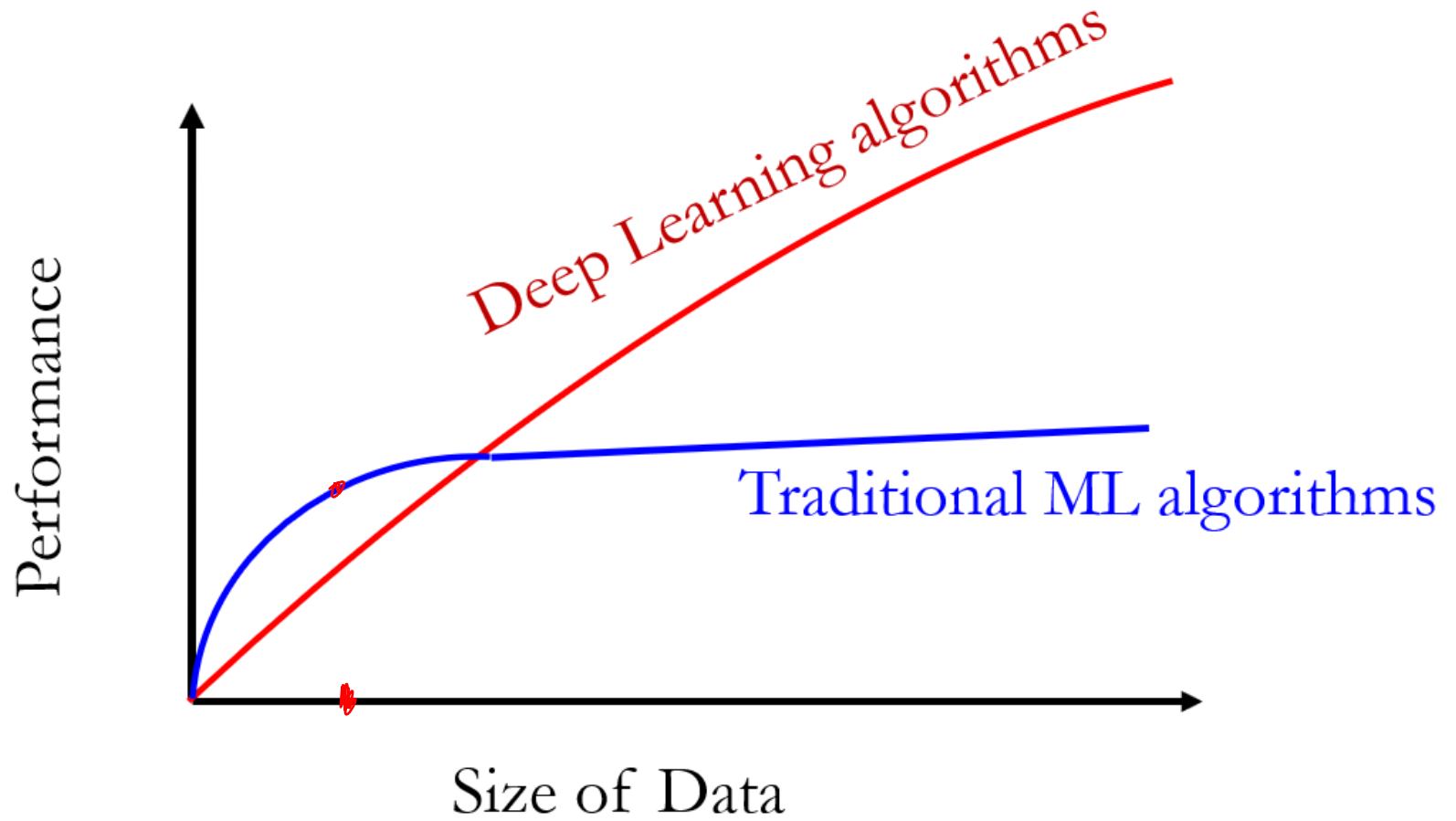
Cost of Computing Power Equal to an iPad 2



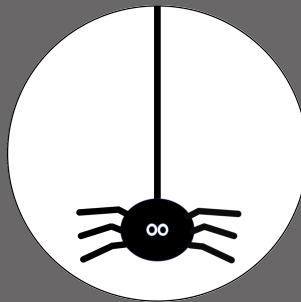
Note: The iPad2 has computing power equal to 1600 million instructions per second (MIPS). Each data point represents the cost of 1600 MIPS of computing power based on the power and price of a specific computing device released that year.

Source: Moravec n.d..

CNN



Neural Network



A linear Model is a Neural Network

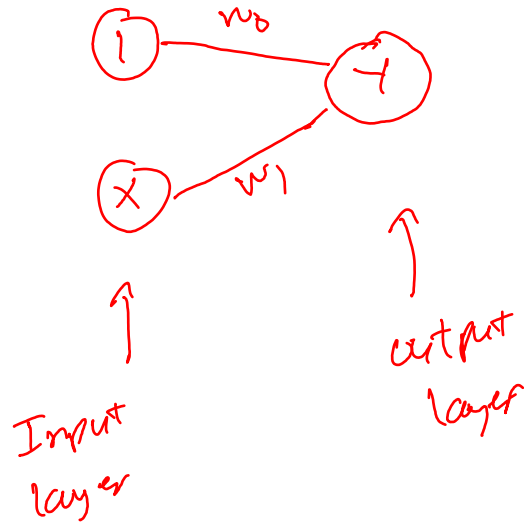
Linear model

$$\begin{array}{c|c} x & y \\ \hline & \end{array}$$

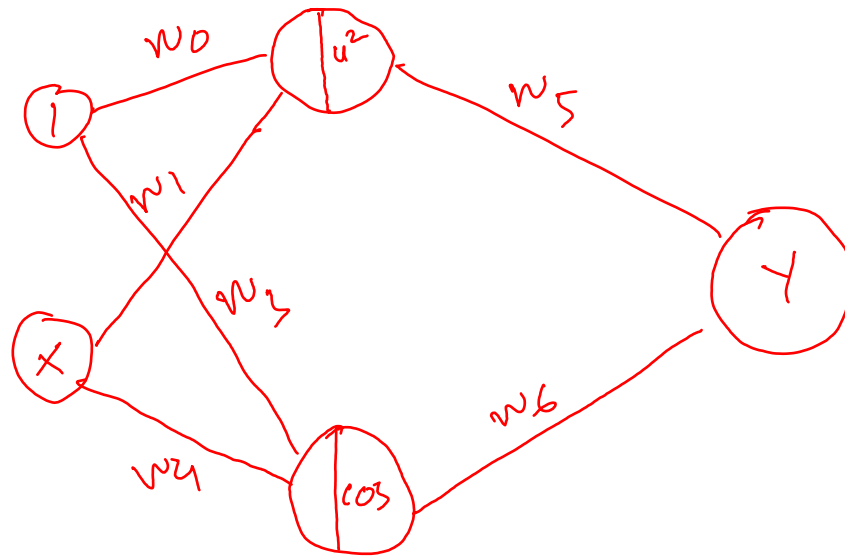
$$y = w_0 + w_1 x \quad (y = a + bx)$$

$$y = w_0 \cdot 1 + w_1 \cdot x$$

2 parameters w_0, w_1



$$y = w_5 (w_0 + w_1 x)^{(2)} + w_6 \cos(w_2 + w_4 x)$$



Input
layer

1 hidden
layer

output
layer

shallow network

parameters:

$w_0, w_1, w_2 \dots w_6$

7 parameters

Deep network would have
many many hidden
layers and parameters

Number of Parameters

Comparison					
Network	Year	Salient Feature	top5 accuracy	Parameters	FLOP
AlexNet	2012	Deeper	84.70%	62M	1.5B
VGGNet	2014	Fixed-size kernels	92.30%	138M	19.6B
Inception	2014	Wider - Parallel kernels	93.30%	6.4M	2B
ResNet-152	2015	Shortcut connections	95.51%	60.3M	11B

- FLOP: Number of Operations

Deeper and Deeper

